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Intelligence as compression

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Problem: Is AI more than a heterogeneous set of clever tools?

Algorithmic information can serve as a theoretical foundation for many aspects of intelligent processing. AIT offered us a definition of unexpectedness and of subjective information. We mentioned Gregory Chaitin's famous sentence: "Comprehension is compression". Let's build on the idea. The following table lists various aspects of intelligent processing and how they can be viewed from a **compressionist perspective**.

We start with statements "X is compression" that are most relevant to the present chapter.

Compression as unexpectedness

Subjective probability is compression	Subjective probability, as measured by unexpectedness, corresponds to the compression between expected complexity and observed complexity. > see this chapter
Aesthetics is compression	Complexity drop often contributes to aesthetics (example: Larry Kagan's wire structures) > see this chapter
Creativity is compression	Actions or innovations are perceived as creative when they elicit surprise, and surprise is analyzed as a complexity drop. > More on this.
Abduction is compression	Abduction is the ability to find out a cause for a problematic state of affairs. The best causal hypotheses, according to Occam's razor, are the simplest ones. Their effect is to diminish the causal complexity of the path to a solution. > More on this.
Relevance is compression	- A relevant statement corresponds to a complexity gap - by pointing to an unexpected situation - by reducing causal complexity through explanation or action - by increasing description complexity through trivialization > More on this.
Responsibility is compression	- The judge decides on responsibility by showing that the defendant's action diminished the causal complexity of the undesirable effect. - Moral judgment also takes the simplicity of the victim (seen from the judge) into account. > More on this.
Emergence is compression	Emergence in collective systems can be defined as the difference between the sum of individual complexities and the collective complexity. > Watch the video at the end of chapter 1.
Humour is compression	Many humorous effects result from unexpected simplicity. > See next exercise.

More on this: visit the

visit →



Spider

0/1 point (ungraded)

-- *Why did the fly fly?*
-- *Because the spider spied her!*

The pun is due to...

- ☐ The causal complexity of the event.
- ☐ The causal complexity of the phonetic similarity between 'spider' and 'spied her'.
- ☐ The unexpected simplicity of the phonetic similarity between 'spider' and 'spied her'.
- ☒ The complexity drop created by the explanation.

Answer
Incorrect:
The pun is due to the unexpected simplicity of the phonetic similarity between 'spider' and 'spied her'.

Submit

You have used 1 of 1 attempt

Now, let’s look at some "X is compression" conjectures that are relevant to computer science in general.

Compression and computer science

Coding is compression	Shannon-Fano codes, Huffman codes, hash codes are methods for designing compact codes. These codes replace the data. Since they are more compact, they achieve compression.
Storing is compression	- Once data is stored, it can be defined by its address. As the latter is generally shorter than the data stored (which might sometimes be arbitrarily large), storing achieves compression. - Indexing methods are used to diminish the average size of addresses.
Forgetting is compression	One should forget only unimportant data. Importance, as relevance, is associated to unexpectedness. As time goes by, description complexity increases (as $\log(t)$), making some data less and less relevant, until they become good candidates for forgetting.
Pattern recognition is compression	Whenever a structure is recognized in an image, the complexity of the pixels or sub-structures it contains drops as they can be predicted from the structure.
Image synthesis is compression	Fixed or animated image synthesis is efficient when it can be controlled using only a few parameters (e.g. recursion depth in a fractal). The fact that realistic images can be compressed down to those few parameters is often remarkable (see the demoscene

	mentioned in Chapter 2).
Visualizing is compression	Large data visualization, for instance for the description of <u>large social networks</u>, requires lossy compression to generate pictures that a human mind can understand.
Schematic is compression	- Schematic results from lossy compression: contours are simplified and some objects are replaced by symbols. Note that the whole point is to keep the semantics and relevance of what is represented. - This is also true of comics.

Hash codes

1/1 point (ungraded)

Some hash code functions such as SHA-256 are designed to be virtually impossible to reverse in practice.
They correspond to lossy compression:

☒ True

☐ False

Answer
Correct:
True, even if Kolmogorov complexity doesn't care about the computational time it would take to determine the input that corresponds to a given output. Compression is lossy because of collisions. When the output of hash function has fixed length while the input may be arbitrarily long, collisions cannot be avoided (even if rare for practical purposes). The indeterminacy of the input means that some information has been lost.

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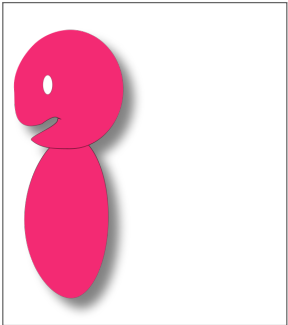
You have used 1 of 1 attempt

Finally, let's look at some other "X is compression" conjectures that are relevant to Artificial Intelligence.

Compression and AI

Information is compression	The smallest program p that generates an object s represents its informational content. It is a compressed version of s. ➤ see <u>Chapter 2</u>.
Learning is compression	This is the whole point of <u>Chapter 4</u>.
Probability is compression	The probability of occurrence of an object is measured by the cumulative probability of the minimal programs that generate it. ➤ see <u>Chapter 3</u>.
Analogy is compression	Good proportional analogies (A is to B as C is to D) minimize the overall complexity of (A, B, C, D). ➤ see <u>Chapter 4</u>.
"Comprehension is compression"	This is Gregory Chaitin's great aphorism. Understanding a phenomenon means replacing observations by a

	theory. A good theory realizes a trade-off between simplicity and accuracy. ➤ More on this.
Proving is compression	• A proof replaces many true statements by a theorem. The theorem is a compressed version of the (often infinite) set of statements it proves. • Conversely, the compression argument can be used in many proofs by contradiction. ➤ see Chapter 3.
Intelligence is compression	• Intelligent behaviour amounts to taking the best rewarding action based on the most probable, <i>i.e.</i> the simplest, future. ➤ see Chapter 4. • And more generally, any intelligent device (and this includes the human brain) seeks to achieving maximal compression (within the constraints of available resources).



That's it.

What did you learn?

Help future students by selecting the skills or subjects you learned from this unit.

- Computer Science

Probability

Artificial Intelligence

Creativity

Forecasting

Data Visualization

Pattern Recognition

Aesthetics

☐ None of these subjects were covered





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